Designing a quiz-based educational dashboard

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We have developed a prototype for a dashboard with an integrated quiz system. The first objective of this dashboard is to help students monitor their progress throughout the semester for them to adapt their workflow. The second objective is to make them aware of their performance on the quizzes to motivate them to work with the course material and improve. The students generate data when taking the quizzes, which is processed into metrics in the form of time spent and accuracy on quizzes, as well as progress. The prototype started out as a few drawn diagrams and was thereafter interated upon in the program Figma. The dashboard allows the students to go from over statistics and drill down into individual courses and quizzes. We have created three scenarios for how a student may want to use the dashboard.

Additional Key Words and Phrases: educational dashboard, educational interfaces, interaction design, learning analytics, quiz, improving performance, workflow management

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1 OBJECTIVES

For students in higher education, doing well at, or at least passing, the final exams in their courses, tends to be the main goal. However, to do well at the exams, most students need to practice the lectured material throughout the semester. Different courses take different approaches to get students to practice the material, such as writing reports, completing assignments, regular tests, and group projects. Balancing work in several courses in an otherwise hectic semester, can be tough with regards to both time management and motivation. Thus, ungraded assignments may quickly end up as chores that are completed and subsequently forgotten about.

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In light of this, we will propose an educational dashboard for students in higher education with an integrated quiz system. The quiz system will let teachers create quizzes to let students practice the material and test themselves. The dashboard collects data from the quiz system, which it visualises. It has the following purpose:

Provide a supportive environment for enhancing students' learning and motivations through self-assessment and progress tracking.

To help students monitor their progress, the dashboard will provide information about upcoming quizzes, their due dates and the status on quizzes in the students's courses that use the quiz system. Progress may also be monitored through the lense of performance, as this tells the student about which subjects and courses they should practice more and which they seem to master. The intention is to make it easy for the students to determine where they should spend their time and what they should prioritise to prepare themselves for the final exams.

As for raising their awareness of their performance, the dashboard will visualise scores, timing and difficulty levels and, to some extent, contrast these with other students' performance. This serves to encourage the students to compete against themselves and against each other to achieve higher quiz scores, creating a lightly gamified learning environment. The visualisations of the dashboard should also encourage the students to retake quizzes to improve their scores. This will let them revisit subjects they struggle with and reinforce what they have learned.

2 AVAILABLE EDUCATIONAL DATA AND METRICS

In this section, it is provided the description of the fictitious educational dataset and the respective metrics used. Considering the objectives described in Section 1, Table 1 below explains the variables in the fictitious educational dataset. For more details on how the features are utilized in the dashboard, please refer to Section 3.

Column Name	Data Type	Description
student_id	int	Each student - with a matriculation number id assigned by the university - has access to the dashboard.
quiz_id	int	The identification number for each quiz inside a course.
timestamp	datetime	The final date by which a student completes a quiz.
task_number	int	The number assigned to each question in a quiz
score	int	The score of a student's answers is weighted based on question difficulty (easy, medium, hard). Each question contributes differently to the overall quiz accuracy score: easy questions add 0.3 points, medium questions add 0.6 points, and hard questions add 1 point, while incorrect answers contribute 0 points.
time	int	The time taken to complete a quiz task, measured in milliseconds.
answer	string	The answer selected by a student for a quiz question.

Table 1. Educational Dataset

The metrics used to develop our dashboard are sourced from the dataset shown in Table 1, which includes primarily student-generated data. While the dataset also contains information such as the course to which each quiz belongs to, here we focus only on student-generated data to provide metrics that support the two objectives described in Section 1. We can outline three main metrics available in the dashboard as follows: *time spent*, *accuracy*, and *progress*.

Time spent

Time spent metrics offer insights into how students allocate time across courses, helping with workload adjustment. In fact, time management is a knwon challenge for students balacing multipule course. By tracking the time spent on quizzes at different levels, students gain insight into their study habits and how they allocate effort across different subjects. The metrics for time spent `[HH:mm]` by a student on the dashboard include:

- Average time spent on quizzes within a single course;
- Time spent on individual quizzes;
- Time spent on the single quiz tasks, categorized by difficulty level (easy, medium, and hard tasks);
- Total time spent on a single course, providing a cumulative view of time in the course page.

Accuracy

Accuracy provides feedback on performance, especially relative to question difficulty, supporting targeted improvement efforts. With accuracy data, students can see where they are meeting their learning goals and how they compare with the other students (e.g. through average class accuracy), creating a motivating environment. The accuracy metrics `[%]` includes:

- Average accuracy across courses, with a comparison between the student's individual average and the class average;
- Accuracy per quiz, based on the student's responses, also categorized by difficulty level (easy, medium, and hard tasks);
- Class accuracy per quiz, presenting anonymized performance data of other students for each quiz;
- Overall accuracy across all completed quizzes for a certain course, including a division by difficulty level.

Progress

Progress shows rates and deadlins, given a structured approach to assignments. Progress tracking is essential for helping students stay on track for deadlines. By displaying the number of completed tasks, this metric allows students to visually see their journay of the courses along the year. The progress metrics include:

- Total quizzes completed per month, with comparisons to both the class average and the student's own average;
- Quiz review summary, providing feedback on each quiz based on the student's responses (correct or incorrect answers);
- Deadline tracking and completion timestamps, allowing students to see quiz deadlines and the dates they finished each quiz;
- Total quizzes completed for each course, displayed on the course page to give an overview of progress within individual courses.

In conclusion, these are the three primary metrics in the dashboard. A detailed explanation of the charts and the journey from the low-fidelity prototype to the final Figma dashboard can be found in Section 3.

3 EDUCATIONAL DASHBOARD(S) PROTOTYPES

The prototyping process helped us conceive and adjust our dashboard according to the objectives we settled on. By first using rough sketches before moving on to a collaborative interface creator in the form of Figma, we were able to test various ideas to figure out what would work best for the students relying on our own experiences. We were able to figure out the main features we would require to reach our objectives.

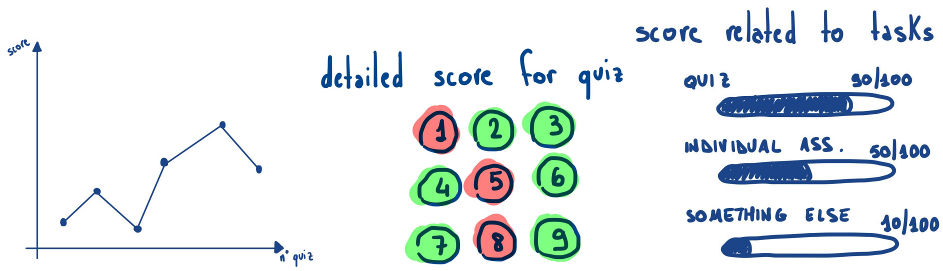

Fig. 1. Sketches of the different features we thought about

To start the prototyping process we sketched out different ideas on paper, as can be seen in figure 1, representing the data we meant to use, trying to see what would be the best display of information. We thought about displaying the evolution of the average score in time, showing a detailed view of the quiz results or showing the average score depending on the type of tasks. Discussing all our ideas and being able to represent them with drawings helped us progressively visualise what our dashboard could look like and enabled us to create a low-fidelity prototype. We made a Figma prototype creating an outline of what the user experience would be, noticing things we had not thought about yet, such as which pages we would use and the different buttons we need for linking them. After having a general idea of the layout of our dashboard, we adjusted the different elements to make the page easy to use so students can easily find what they need and make it more intuitive. We chose to go with a simple design with few elements in color to make them stand out more and show what is important for the student to notice. Thanks to this prototyping we were able to point out the main features we needed and make a good user experience to encourage and motivate students while showing them what they need to focus on.

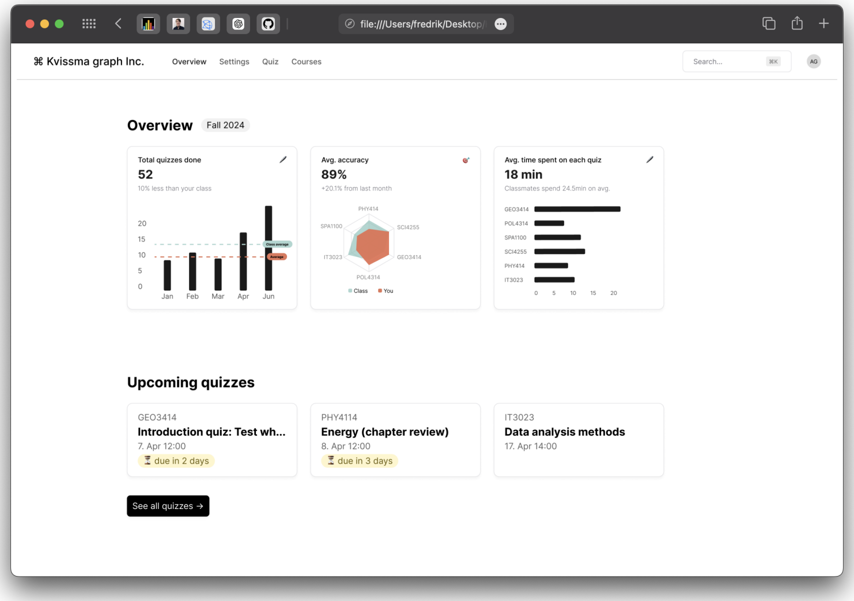

Fig. 2. Home page of our final dashboard

On the dashboard's home page, shown in figure 2, we decided to display an overview of different statistics to show the student where they are at and what they should focus on. First we show them how many quizzes they have completed this semester and whether they are above or below average compared to the rest of the class. This will show their progress and help motivate them to catch up, as well as showing progress through the semester.

We then have a spider-web diagram that shows a student's accuracy in each enrolled course and compares it with the average from the rest of the class. This helps the students visualize which courses they need to give more attention.

Finally we show the average time spent on a quiz for a given course, and then compares it with the other courses. This helps visualizing how much time is spent on each course and which courses tend to feature the most demanding quizzes.

Underneath this overview we show the students the upcoming quizzes that need to be completed next, with the quizzes' due times. Quizzes being due the coming two days are marked as urgent with a yellow pill.

From this home page, students also have access to the settings, the list of available and completed tests and the list of current and previous courses.

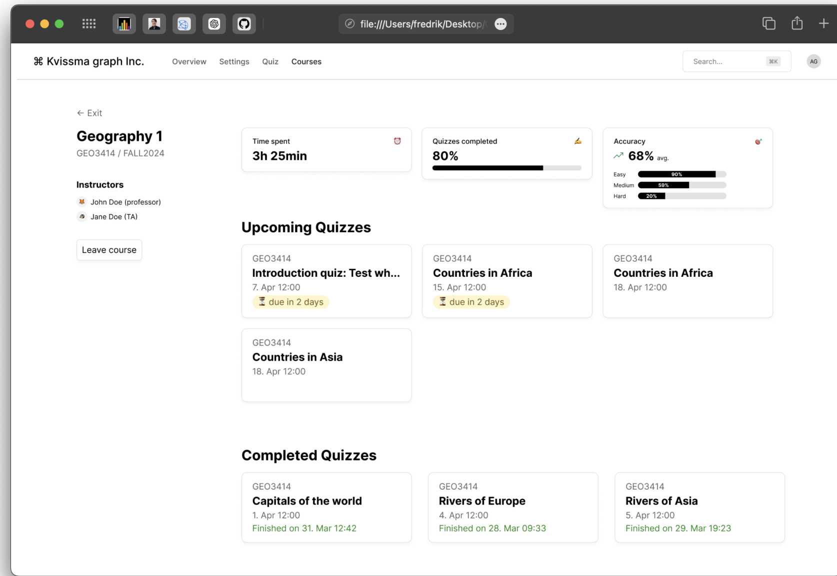

Fig. 3. Course overview page

If a student clicks on one of the courses they will arrive on the Course overview page shown in Figure 3. On this page the student can see the instuctors of the class and the code of the course. They will be able to see the time spent on this course's quizzes and the number of quizzes completed. Total accuracy for all the quizzes in the course and accuracy distributed accross difficulty levels is also visualised. From this page, the student can access the upcoming quizzes for the course as well as the completed quizzes to go back on them and see a detailed view of their results on them to know what they still has to learn.

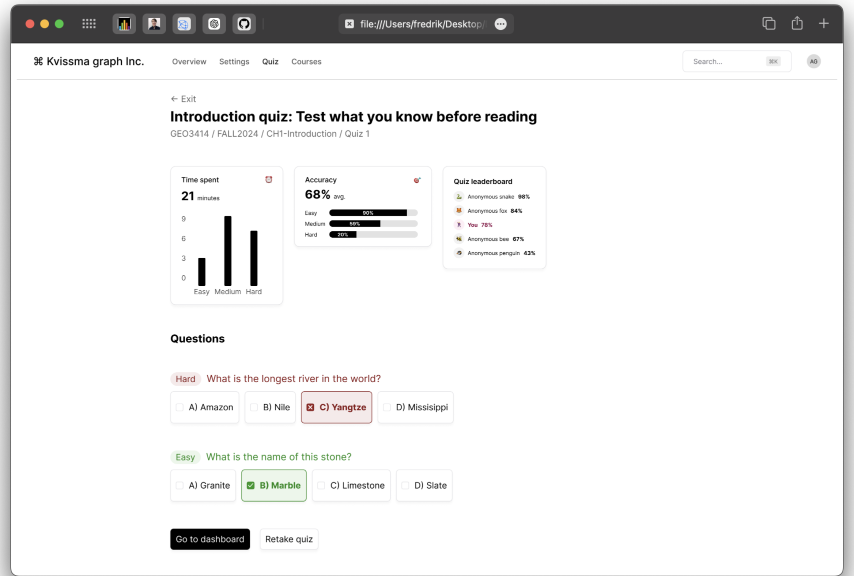

Fig. 4. Page for the detailed results of a quiz

The detailed results of a quiz enable students to go back onto their mistakes to figure out what they missed on the lecture materials. We also show the student his score depending on the difficulty of the question as well as the time spent on a question based on its difficulty. We also added a leaderboard showing where the student is compared to others to motivate him to retake the test and score better.

4 STORY-BOARDING WITH RESPECT TO THE PROPOSED EDUCATIONAL DASHBOARD(S)

4.1 Scenario 1: Doing a quiz to improve your statistics

Here is the most common scenario, shown in Figure 5, where a student goes for logging in and looking at the overview page to see their current statistics. Because they already wanted to or because their statisics influenced them, they then go for the first recommended quiz, answer every question and end the quiz. After ending the quiz, they take a look at the detailed results of their session. Finally, they can either go back to the overview page with the statistics actualised or if they are not satisfied with their results on the quiz, they can also retake it. If they go back to the overview page, the newly taken quiz will disappear, and the numbers and diagrams will update to include data from this quiz. If other students do quizzes in some of the same courses as the student is enroled in, the blue line for average number of quizzes and blue accuracy polygon will change, as well.

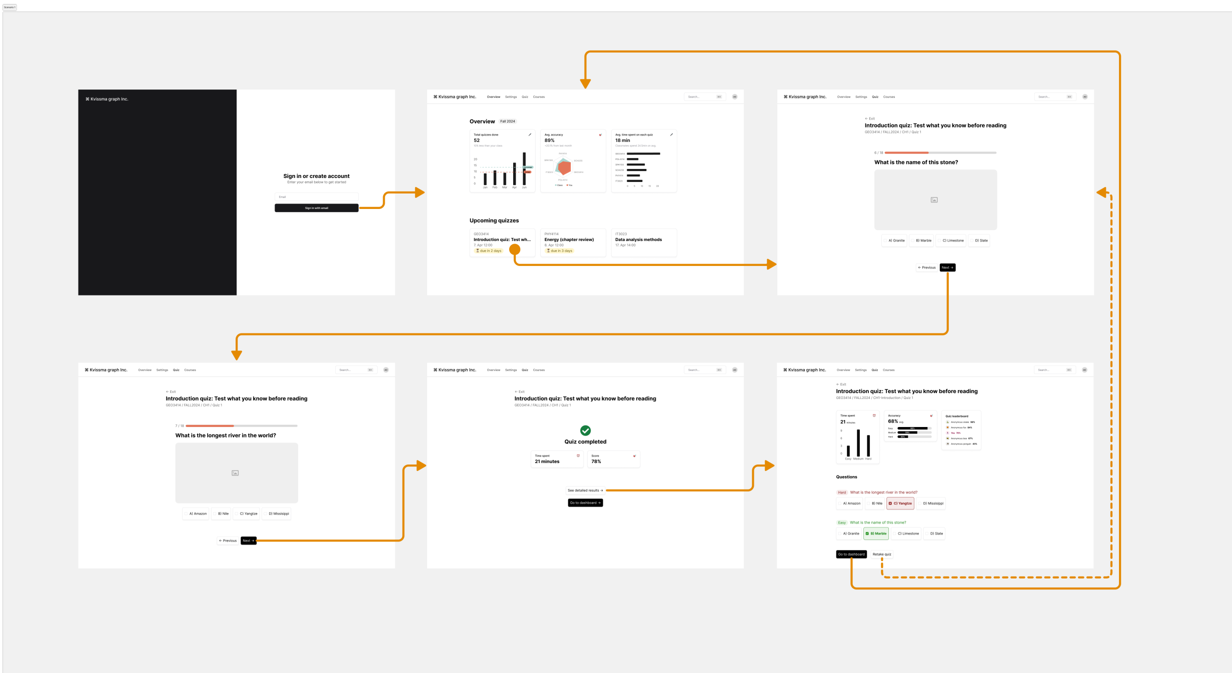

Fig. 5. Scenario 1: A student does a quiz to improve their statistics

4.2 Scenario 2: Improving your score by re-taking a previously completed quiz

This scenario in Figure 6 looks into how a user can engage with the quizzes and improve their scores. It shows how the user can go from the main dashboard page and into the "Quiz" page, which shows a list of the upcoming and completed quizzes. To retake a quiz, the user must click on a completed quiz to view the detailed results, and then press the "retake-button" on the bottom. The quiz will then start from the beginning, and the user can see their updated results after completion.

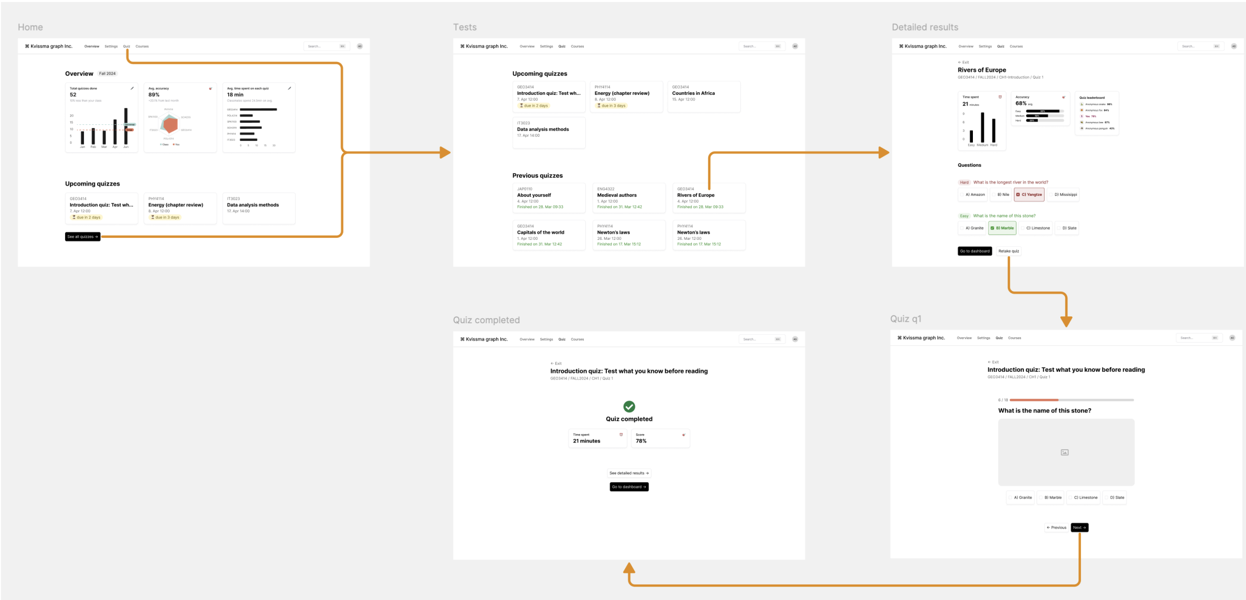

Fig. 6. Scenario 2: A student improves their score by re-taking a previously completed quiz

4.3 Scenario 3: Checking statistics for a specific course

Students usually take more than one course a semester, and switching between work in different courses incures a context switch that can lead to wasted time and students not entering a good flow in the work. The dashboard, therefore, ensures that quizzes are connected to the students' different courses. This makes it possible for the student to see statistics about specific courses and to compare progress and performance in their courses.

Scenario 3, depicted in Figure 7, shows a situation where a student is curious about how they are doing in a specific course - Geography 1 in this case. The student starts out at the overview page, where they can compare their accuracy on quizzes in the different courses with that of the class average. Here they can also see how much time they spend on average per quiz in the various courses. This can inform them of which courses they spend the most time in and which courses figure the most demanding quizzes.

The student then continues to the "courses" page, where they see an overview of all their current and previous courses. Here they notice that Geography 1 has four unfinished quizzes, which can mean that this is the course where they are the most behind on work. To see statistics and the quizzes for this course, they click on it. On the course details page, they can guage their progress through see the upcoming quizzes and through the "Quiz completed" progress bar. They can also see how much time as gone into doing quizzes in this course, as well as their accuracy. Accuracy is also shown for the different difficulty levels.

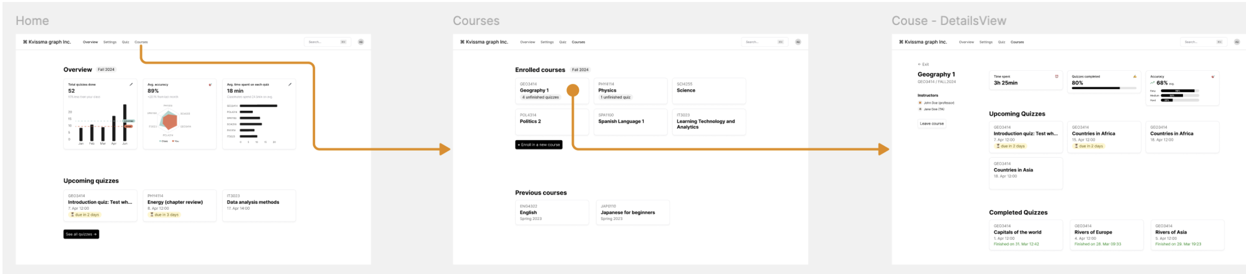

Fig. 7. Scenario 3: A student checks their statistics for a specific course.

CONTRIBUTIONS STATEMENT

See table 2 for the distribution of our contributions to this report and the making of the dashboard prototype.

Table 2. Contribution statement

Contribution	**Names**
Discussing the objectives	Everyone through group meetings
Initial diagram sketches	Peiretti Riccardo
Setting up Figma work space, acquiring Figma assets and creating Figma prototype from the design	Fredrik Fonn Hansen
Designing the dashboard in Figma	Everyone through group meetings and collaborative work in Figma
Writing the Objectives part	Sander Østrem Fagernes
Writing about the data and metrics	Peiretti Riccardo
Writing the Prototypes part	Manon Debled and Clément boudier
Planning and setting up storyboards in Figma	Fredrik Fonn Hansen,Sander Østrem Fagernes and Tao Durand-Vizcaino
Writing the story boarding part	Fredrik Fonn Hansen,Sander Østrem Fagernes and Tao Durand-Vizcaino

APPENDIX

A Figma Prototype

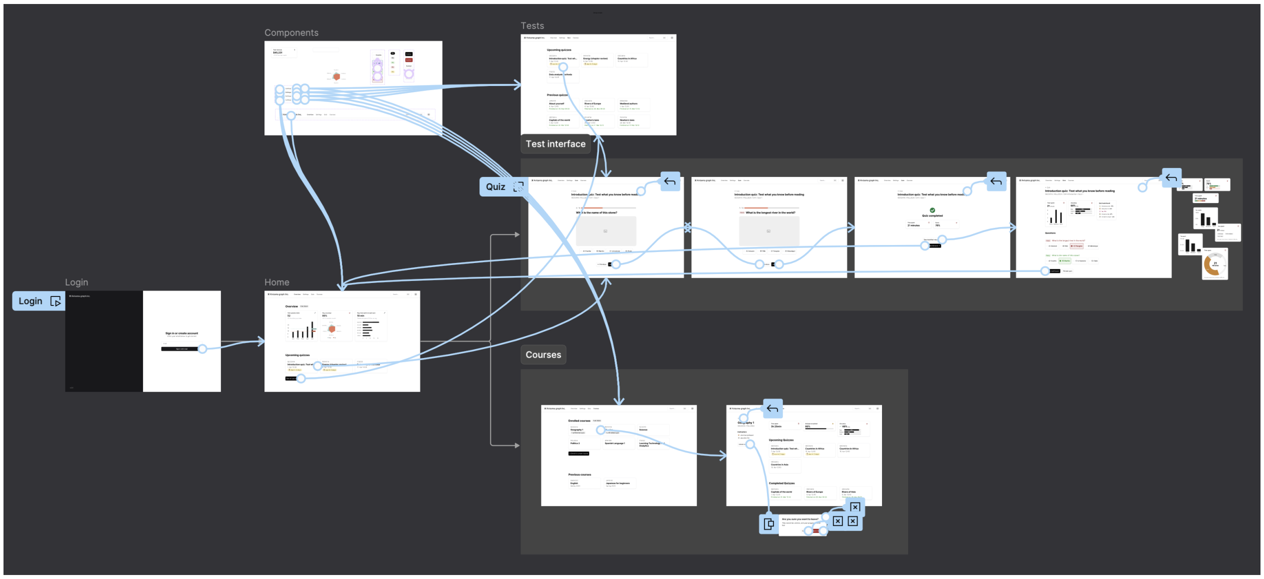

Fig. 8. Our final Figma Prototype